

HW2 STAT 325

A) $Y_{ij} = \mu + \tau_i + \epsilon_{ij}, i=1, \dots, a, j=1, \dots, n$

① $E(Y_{ij}) = E(\underbrace{\mu + \tau_i}_{\text{constant}} + \underbrace{\epsilon_{ij}}_{\epsilon_{ij} \sim N(0, \sigma_{\epsilon}^2)}) = \mu + \tau_i$

$Var(Y_{ij}) = Var(\underbrace{\mu + \tau_i}_{\text{constant}} + \epsilon_{ij}) = Var(\epsilon_{ij}) = \sigma_{\epsilon}^2$

thus $Y_{ij} \sim N(\mu + \tau_i, \sigma_{\epsilon}^2)$

② $\bar{Y}_{i.} = \frac{1}{n} \sum_{j=1}^n Y_{ij} = \frac{1}{n} \sum_{j=1}^n (\mu + \tau_i + \epsilon_{ij})$
 $= \frac{1}{n} (\underbrace{n\mu}_{\text{constant}} + \sum_{j=1}^n \tau_i + \epsilon_{ij})$

$E(\bar{Y}_{i.}) = \mu + E(\sum_{j=1}^n (\tau_i + \epsilon_{ij})) = \mu + \tau_i$
 $= \mu + E(\tau_i) + E(\sum_{j=1}^n \epsilon_{ij}) = \mu + \tau_i$
 $Var(\bar{Y}_{i.}) = Var(\sum_{j=1}^n (\mu + \tau_i + \epsilon_{ij})) = Var(\sum_{j=1}^n \epsilon_{ij}) = \frac{\sigma_{\epsilon}^2}{n}$

$E(\bar{Y}_{..}) = E(\sum_{i=1}^a \sum_{j=1}^n \mu + \tau_i + \epsilon_{ij}) = \mu + \frac{1}{an} \sum_{i=1}^a (E(\tau_i) + \sum_{j=1}^n E(\epsilon_{ij}))$
 $= \mu + \frac{1}{an} E(\sum_{i=1}^a \tau_i) + E(\sum_{j=1}^n \epsilon_{ij})$
 $= \mu + \frac{1}{an} E(0) + 0 = \mu$

$Var(\bar{Y}_{..}) = Var(\sum_{i=1}^a \sum_{j=1}^n \mu + \tau_i + \epsilon_{ij}) (\frac{1}{an})^2$

continue

$$\frac{1}{(an)^2} \text{Var} \left\{ na\mu + \sum_{i=1}^a \sum_{j=1}^n \tau_i + \varepsilon_{ij} \right\}; \quad \text{Var}(na\mu) = 0$$

$$= \frac{1}{(an)^2} \text{Var} \left[\sum_{i=1}^a \sum_{j=1}^n \tau_i + \varepsilon_{ij} \right]$$

$$= \frac{1}{(an)^2} \text{Var} \left[\sum_{i=1}^a \left(\tau_i + \sum_{j=1}^n \varepsilon_{ij} \right) \right]; \quad \sum_{i=1}^a \tau_i = 0.$$

$$= \frac{1}{(an)^2} \text{Var} \left[\sum_{i=1}^a \sum_{j=1}^n \varepsilon_{ij} \right] \quad \text{Since } \text{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$$

$$\forall i, j \in [1, a] \times [1, n]$$

$$= \frac{1}{(an)^2} \text{Var}(\sigma_\varepsilon^2) = \frac{\sigma_\varepsilon^2}{an}$$

$$(B) \quad S_y^2 = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 \quad \text{and} \quad P_y^2 = \sum_{j=1}^n (y_{.j} - \bar{y}_{.})^2$$

We have

$$E(S_y^2) = E \left(\sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 \right) = E \left(\sum_{i=1}^a \sum_{j=1}^n (y_{ij}^2 - 2y_{ij}\bar{y}_{i.} + \bar{y}_{i.}^2) \right)$$

$$= \frac{1}{N-1} \sum_{i=1}^a \sum_{j=1}^n \left\{ E(y_{ij}^2) - 2E(y_{ij}\bar{y}_{i.}) + E(\bar{y}_{i.}^2) \right\}$$

$$= \frac{1}{N-1} \sum_{i=1}^a \left(\sum_{j=1}^n \sigma_\varepsilon^2 - n\bar{y}_{i.}^2 \right) = \frac{1}{N-1} E \left(\sum_{i=1}^a \sum_{j=1}^n \varepsilon_{ij}^2 - \sum_{i=1}^a n\bar{y}_{i.}^2 \right)$$

$$= an\sigma_\varepsilon^2 - a\sigma_\varepsilon^2 = a\sigma_\varepsilon^2(N-1) = \sigma_\varepsilon^2$$

$$E(P_y^2) = \frac{1}{n-1} \sum_{j=1}^n (y_{.j} - \bar{y}_{.})^2 = \frac{a(N-1)}{n-1} \sigma_\varepsilon^2 = \sigma_\varepsilon^2$$

S_y^2 is UE for σ_ε^2
 S_y^2 is UE for σ_ε^2 by theorem.
 P_y^2 is UE for σ_ε^2 ✓

$$(C) SS_{Total} = (N-1)S_y^2, \quad MS_{Total} = \frac{SS_{Total}}{N-1}$$

$$SS_{Error} = \sum_{i=1}^a (n-1)S_i^2 \text{ and } MS_{Error} = \frac{SS_{Error}}{N-a}$$

$$E(MS_{Total}) = \sigma_e^2 \text{ with } E\left(\frac{(N-1)}{(N-1)} S_y^2\right) = \sigma_e^2$$

$$E(S_y^2) = \sigma_e^2 \text{ shown previously in } a(n-1) \text{ and } n-a = N-a$$

$$(B) \text{ Same for } E(MS_{Error}) = E\left(\frac{1}{\sum_{i=1}^a (n-1)} S_i^2\right)$$

$$= E\left(\frac{(n-1)a}{N-a} \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2\right)$$

$$= E\left(\frac{(n-1)a}{na-a} \sigma_e^2\right) = E\left(\frac{(n-1)a}{a(n-1)} \sigma_e^2\right) = \sigma_e^2$$

$$(D) E(MS_{Error}) = \sigma_e^2 \text{ regardless of}$$

$$\sum_{j=1}^n y_{ij} \text{ are same or different.}$$

$$E(MS_{Total}) = \sigma_e^2 \text{ only if } \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 = 0$$

which means the all effects are the same. Also they have different degree of freedom.

$$(E) SS_{Total} = (N-1) \sum_i \sum_j (y_{ij} - \bar{y}_{..})^2$$

$$= \sum_i \sum_j (y_{ij}^2 + \bar{y}_{..}^2 - 2y_{ij}\bar{y}_{..}) = \sum_i \sum_j y_{ij}^2 + \sum_i \sum_j \bar{y}_{..}^2 - 2 \sum_i \sum_j y_{ij}\bar{y}_{..}$$

$$\sum_i \sum_j x_{ij}^2 + \underbrace{\sum_i n x_{i.}^2}_{n \sum_i x_{i.}^2} - 2 \sum_i \sum_j x_{ij} \bar{x}_{i.}$$

Q5 total

$$\textcircled{E} \sum_i \sum_j (x_{ij} - \bar{x}_{..})^2 = \sum_i \sum_j (x_{ij} - \frac{\bar{x}_{..}}{N})^2$$

$$= \sum_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n}} (x_{ij})^2 + \sum_{1 \leq i \leq n} \sum_{1 \leq j \leq n} \left(\frac{\bar{x}_{..}}{N} \right)^2 - 2 \sum_i \sum_j x_{ij} \frac{\bar{x}_{..}}{N}$$

$$= \sum_i \sum_j x_{ij}^2 + n \cancel{\bar{x}_{..}} \frac{\bar{x}_{..}^2}{N} - 2 \bar{x}_{..} \bar{x}_{..}$$

$$= \sum_i \sum_j x_{ij}^2 + \frac{\bar{x}_{..}^2}{N} - 2 \frac{\bar{x}_{..}^2}{N} = \sum_i \sum_j x_{ij}^2 - \frac{\bar{x}_{..}^2}{N}$$